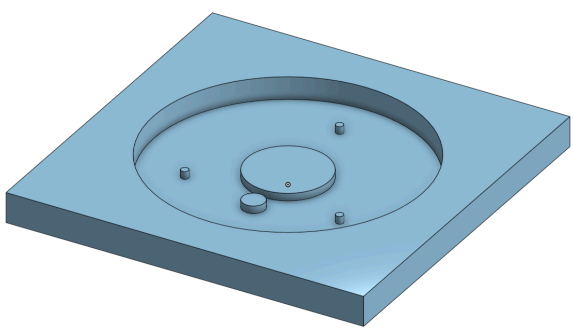
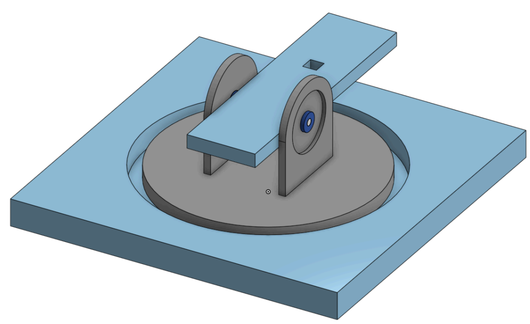
Element G - Jay

After deciding upon the mechanisms for horizontal and vertical traverse, we began thinking of how to construct a basic prototype. Attempting a full scale prototype would be unfeasible, time consuming, and potentially create expenses that would cause us to go over budget when creating the full project. To circumvent this problem, it was decided to create a basic prototype by scaling down a lot of the ideas that were to be used in the full scale prototype. To create it, we planned to use a combination of CNC created parts, and 3D printed parts, as well as spare stock material for a base. This would allow us to test the horizontal mechanism, as well as the vertical mechanism using parts created accurately and without much trouble. Following this conclusion, designing prototypes began on Onshape. The very first idea came out as shown below.



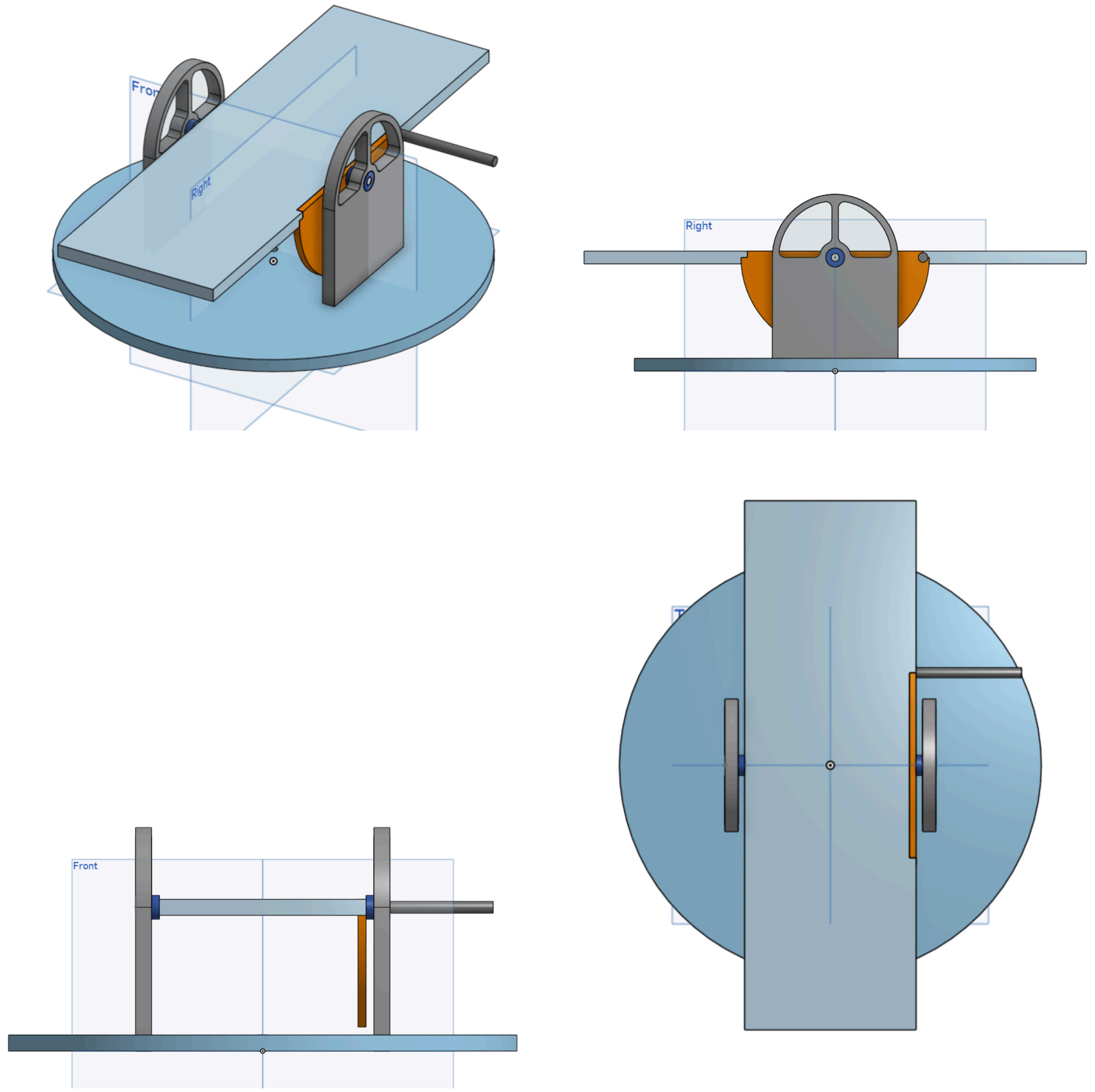


Left: the base that the prototype would be made on, along with nylon spacers and two spur gears. Right: the base with the rotating and elevating platforms attached. The dark blue circles are spacers holding a thin rod in place, which travels through the elevating platform to give it free motion.

This idea originally was planned to test everything upon the creation of the project, this prototype idea was ultimately dropped for a multitude of reasons.

1. The azimuth mechanism was planned to be a ring gear (which the elevation mechanism would be based off of) to create redundancy in the future. The idea drawn above used two spur gears, which would not test what we wanted.
2. The scale at which it was drawn would allow for the prototype to be fast to assemble, but raised significant doubts over fitting the motors to test mechanization would be possible (this also goes back to reason 1; placing a motor underneath the base would require too large of a platform to be made).
3. The elevation mechanism's partial ring gear had no place to anchor onto the elevating platform, and thus could cause problems with proper rotation; if the ring gear's center was not centered with the axle of rotation, the misalignment would cause the elevating platform to raise unevenly, and face the torque (imposed by the gear spinning the ring gear) in unexpected ways, translational force.

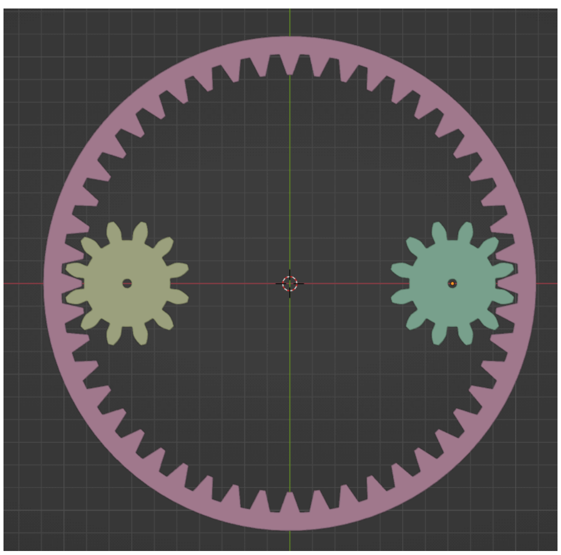
Subsequently, a second prototype was soon drawn, but with separate stages to avoid some previously mentioned problems.

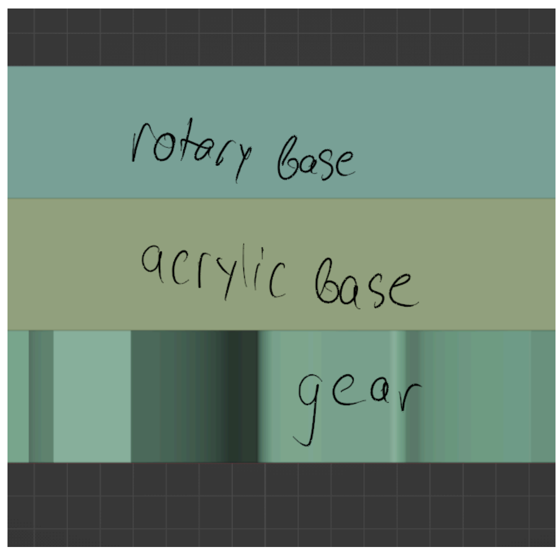


This idea didn't have a proper base modeled in CAD, as it would end up an acrylic sheet with holes drilled in. The holes in the frames holding the elevating base in place were added to reduce weight (though they most likely could have been outright removed). A ring gear was reintroduced for the horizontal mechanism (explained further below), and for the original drawing, the elevation mechanism was replaced with a handlebar and semicircle. The first "stage" of this prototype did not include any motors, as to ensure the horizontal mechanism's gears functioned properly, as well as ensuring a ring gear could be placed in lieu of the orange semicircle, by ensuring the orange semicircle stayed consistently centered with the rod passing through the elevating base. If this was unchanged, then the partial ring gear could replace the orange semicircle, only that the ends of the ring gear would be connected by the notches used to hold the orange semicircle in place.

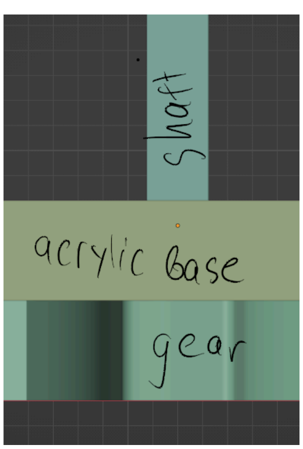
Addressing the problems brought up with the first design:

1. The ring gear was eventually implemented with the image shown below. This would be connected to the rotary base through the acrylic base with a small rod, using friction to keep the two together (this was only possible with the small scale prototype, a fact realized later.)



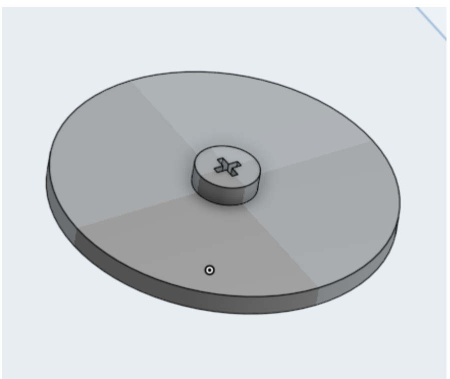


2. The base was larger, measuring 8 inches wide. Using this larger scale allowed for a motor to be used later on to test the elevation mechanism of the base. Having the powered gear and the rotary base's gear be so far apart (because of the ring gear) also allowed for a motor to be connected through a hole in the acrylic base.



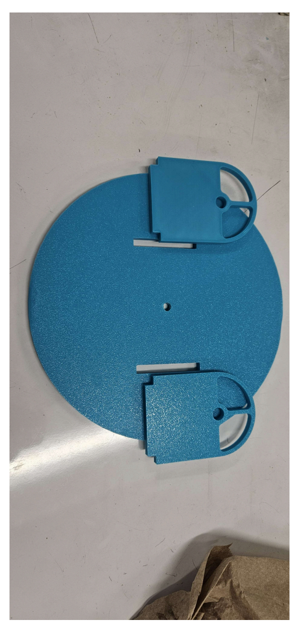
3. For the first stage, this problem was unconsidered, as the orange semicircle took the partial ring gear's place. Eventually, with the second stage, this was no longer a factor, as the partial ring gear would anchor to the elevating platform the same way the orange semicircle would: with the small notches cut into the elevating base.

The second stage would implement motors to confirm the systems used could be motorized without much problems. The rod connecting the powered gear for the horizontal

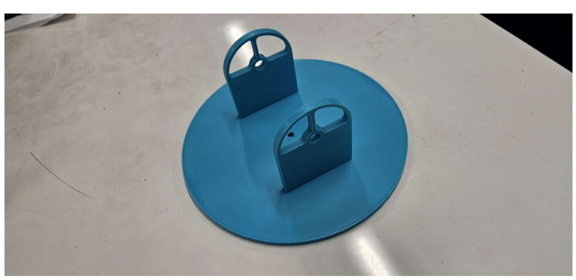


mechanism was originally a shaft with a head. The head was composed of a small circle that could be hand spun, and had a standardized screw head atop, if we so desired to spin the non motorized base with a drill. This piece would be outright replaced with the motor's shaft once motorized, and a small notch on the motor's shaft would be used to hold the motor's shaft in sync with the motorized gear. The elevation mechanism would do the same, with its motorized gear also having a motor plug into it.

This prototype would be unable to consider how the motors would perform with the total weight of the full scale prototype, as it was much smaller. As such, the motors would have to be held by hand, as reasonably sized supports for the components could not be made without cumbersome size. We eventually decided to go with this, dubbing it P1 for future reference.

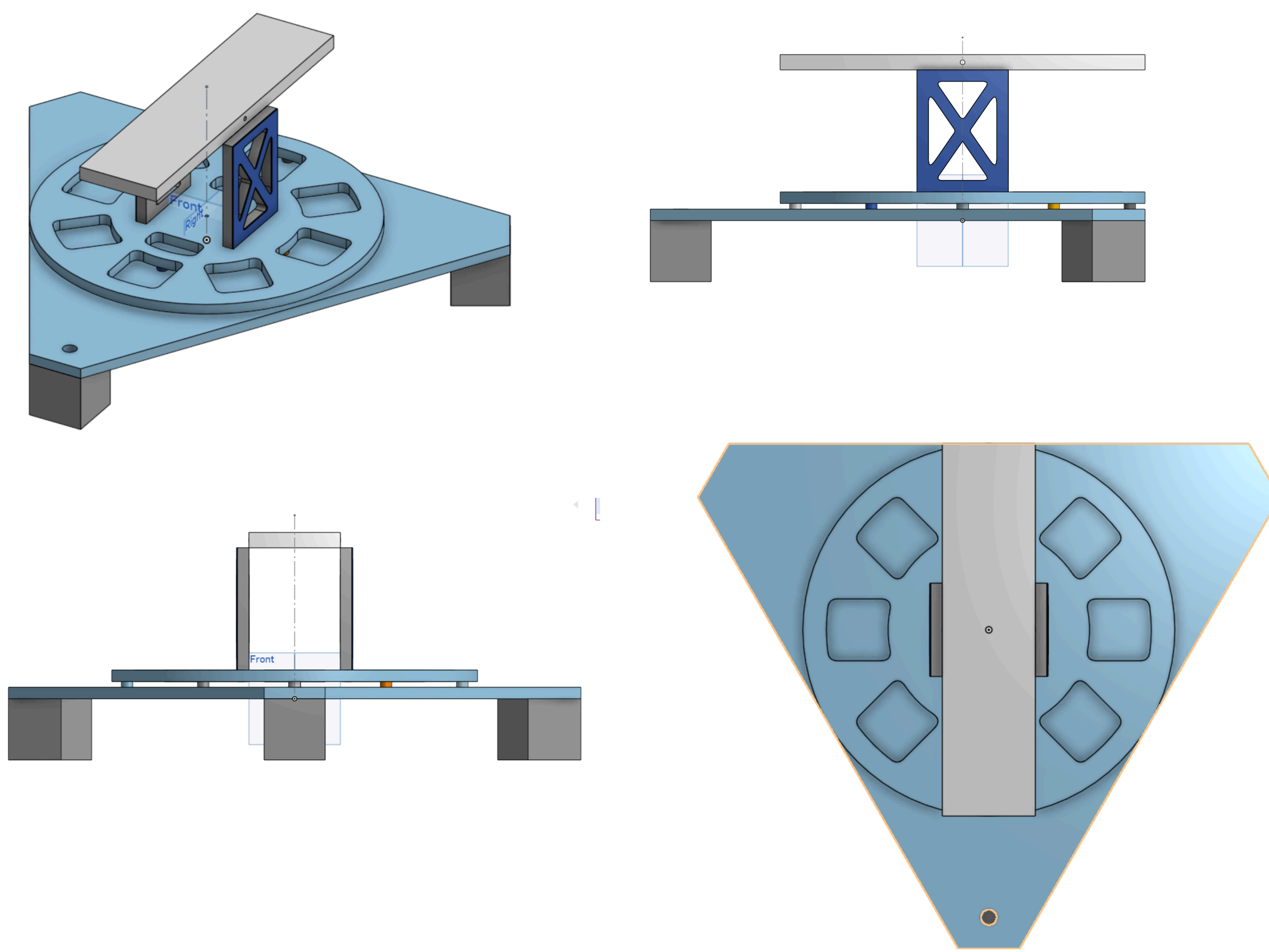


Stage 1 was performed for the horizontal base at

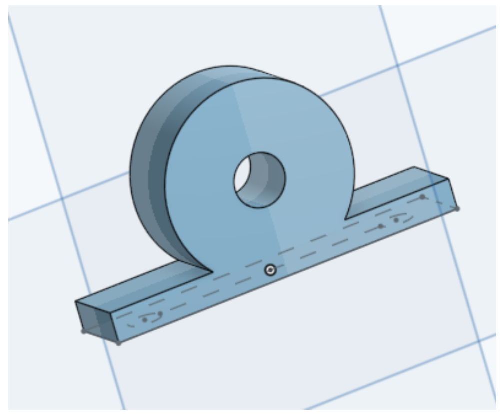


first, using 3d printed parts to create the base. The gears were made separately, and originally the rotation of each was tested by spinning the spur gear in the ring gear (as seen many times in class). This all was placed on the previous mentioned sheet of acrylic, where the ring gear failed due to a simple reason: balance. The two spur gears were not properly spaced under the acrylic sheet, thus they were not on opposite sides of the ring gear. As a result, the ring gear did not work nearly as efficiently as possible, and thus considering it was dropped by P2. The elevation mechanism was unable to be tested by the time we were required to present, and eventually would ultimately be scrapped due to time constraints.

Full scale prototyping would begin with P2.

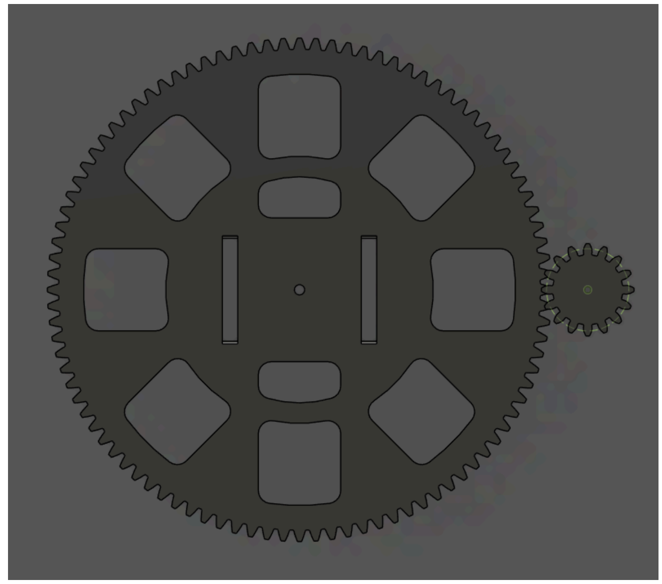


Overall, holes were introduced throughout the rotary base and frame in an attempt to reduce the weight as much as possible while avoiding reducing too much structural integrity. This design dropped the extra braces atop the elevating platform's frames, and instead aimed for a thin sheet of aluminum (shown in dark blue) placed adjacent to the surface of the frames, acting as an extra layer of support. A ball bearing would be used to hold the rod passing through the elevating platform, simplified in CAD as shown to the right. These would be placed atop the frames, and the smaller screw inside the bearing would be used to secure the rod in place.

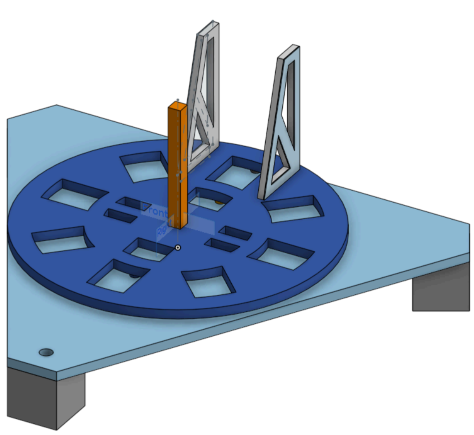


Much heavier materials were proposed for this prototype due to the uncertainty of the weight of the launcher team's group, calling for the rotary base and frames to be made of 0.75" MDF board. The base would be made of a 0.5" sheet of plywood, with nylon spacers separating it from the rotary base. The rotation of the base would be provided by a lazy susan, with the connection between the rotary base and the spur gear in the horizontal mechanism being a 0.5" aluminum rod. A problem here was realized in the balance of such a system, as it would entail the aluminum rod be exactly in the center of the rotary base, which would already be hard to predict with the lazy susan's holes. This aluminum rod also had no reasonable way to ensure the rotation of the rotary base was in sync with that of the spur gear. Another problem was realized the sheer weight, as the motors would have most likely struggled handling such heavy systems. This would also be worsened by the fact that the ring gear's spur gears were 1:1, meaning no torque would be relieved from the motor.

These problems combined with the introduction of the propulsion team's air cannon, P2 slowly was readjusted into P3, and eventually P4. P2 also faced the changing of the azimuth system from a ring gear to a basic spur gear, the teeth of which would surround the rotary base. Instead of having the rotational energy of the motorized gear be transferred through a ring gear and eventually a spur gear rotating in tandem with the rotary gear, the rotary base would also be the gear, and have torque exerted directly upon it, also taking advantage of a more favorable gear ratio of 6:1. The teeth surrounding each base would eventually have to be added later after designing in Fusion 360, as Onshape had no widely available yet efficient tool to create a wide variety of gears.

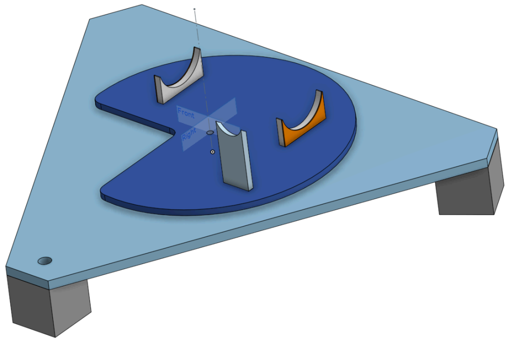


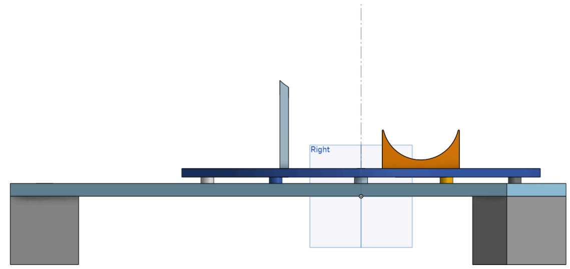
P3 saw a very short life, attempting to take advantage of how torque increases the further the torque is applied from the axis of rotation.

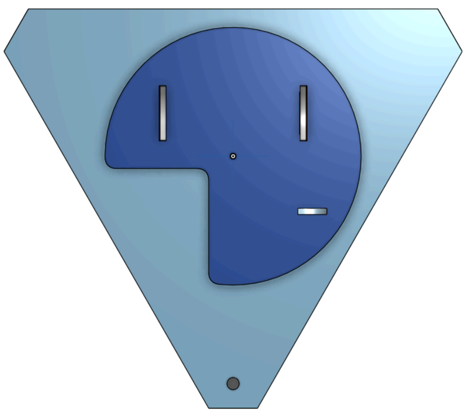


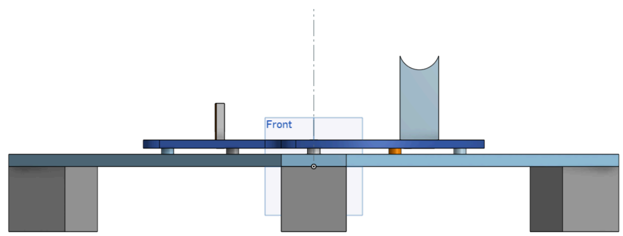
It aimed to let the vertical mechanism have the partial ring gear be on the opposite end of the base from the axis of rotation to maximize torque. A problem arose from this, however, as the weight of the elevating platform would become a major factor in the torque of the system. To counteract this, an orange support beam was erected in the center of the base to hold the elevating platform up when at rest, greatly reducing the strain and increasing the life span of the motor. As time slowly ran out, the feasibility of completing an elevation mechanism that had not been tested prior grew dim. This fact combined with the feedback from Kris Schneider caused P3 to end, and P4 would be created, completely removing the elevation mechanism.

P4 outright removed the elevation mechanism, and replaced the heavy 0.75” MDF boards with lighter 0.5” plywood boards. The rotary base was smaller, and a cutout was made in the corner to reduce weight. The lack of teeth at this portion of the rotary base would not be a problem, as the rotary base was never planned to spin a full 360 degrees. The frames now held up the air canister of the air cannon, and the barrel. The air canister’s frames retained the aluminum braces introduced in P2 (and retained in P3), while a bevel was applied to the air cannon’s barrel’s frame, such that it would angle the cannon up at a fixed 30 degrees. These frames were made with specifications from the launcher’s team, such that it would (assuming perfect tolerances) perfectly hold the barrel and air canister atop it. Clearly, zero tolerance machining was impossible at our scale, thus some leeway was added into the cuts.





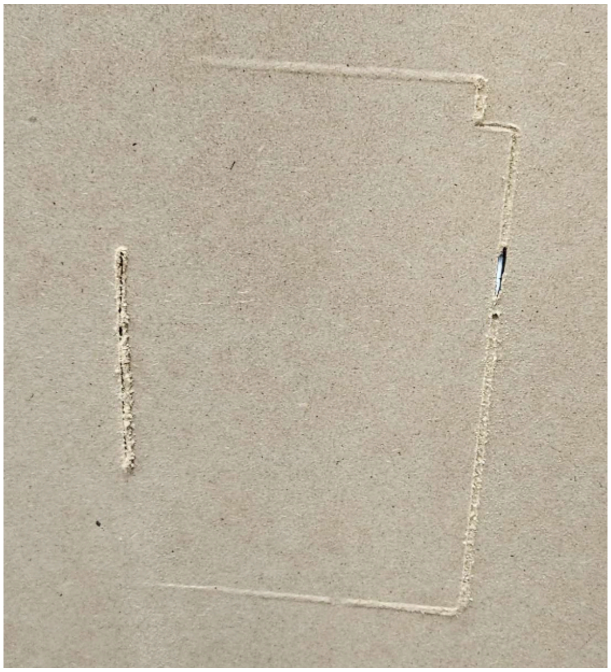


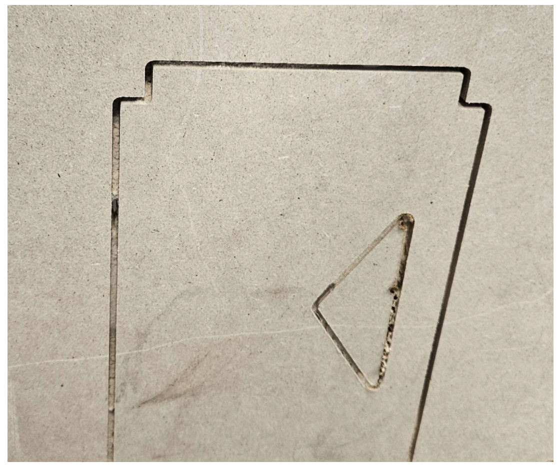


No prototypes from P2 to P4 were ever fully constructed due to problems with the CNC. Wobbling involved with the router of the CNC turned out to cause inconsistencies:

1. The bird's eye view of more complex shapes cut by the CNC would be inconsistent, as seen with attempts to cut small spur gears used for testing; the teeth were all different shapes, and none were uniform enough to properly test with.
2. Oddly, the wobbling of the router led to the CNC to not cut consistently in the vertical axis. The image on the right shows the top view of the MDF board used to try and cut the

frames for P2, and the image on the left shows inconsistent cutting involved with the bottom layer of the same cut.





3. The CNC became misaligned over time, and combined with overheating, caused disastrous results when attempting to cut P2's rotary base. On the final layer, the router became misaligned with its previous paths, and eventually burned the MDF, ruining both the cut and the router bit used to cut out the rotary base. This problem was prevalent even on smaller cuts: when attempting to cut the acrylic circle for Barnett's puzzle feeder, the CNC didn't originally cut all the way through the sheet. So, we attempted to cut the same path, but only this time lowering the height of the router bit beforehand, such that it would cut deeper. What resulted though, was the CNC following the same path, but misaligned in the X-Y axis somehow.